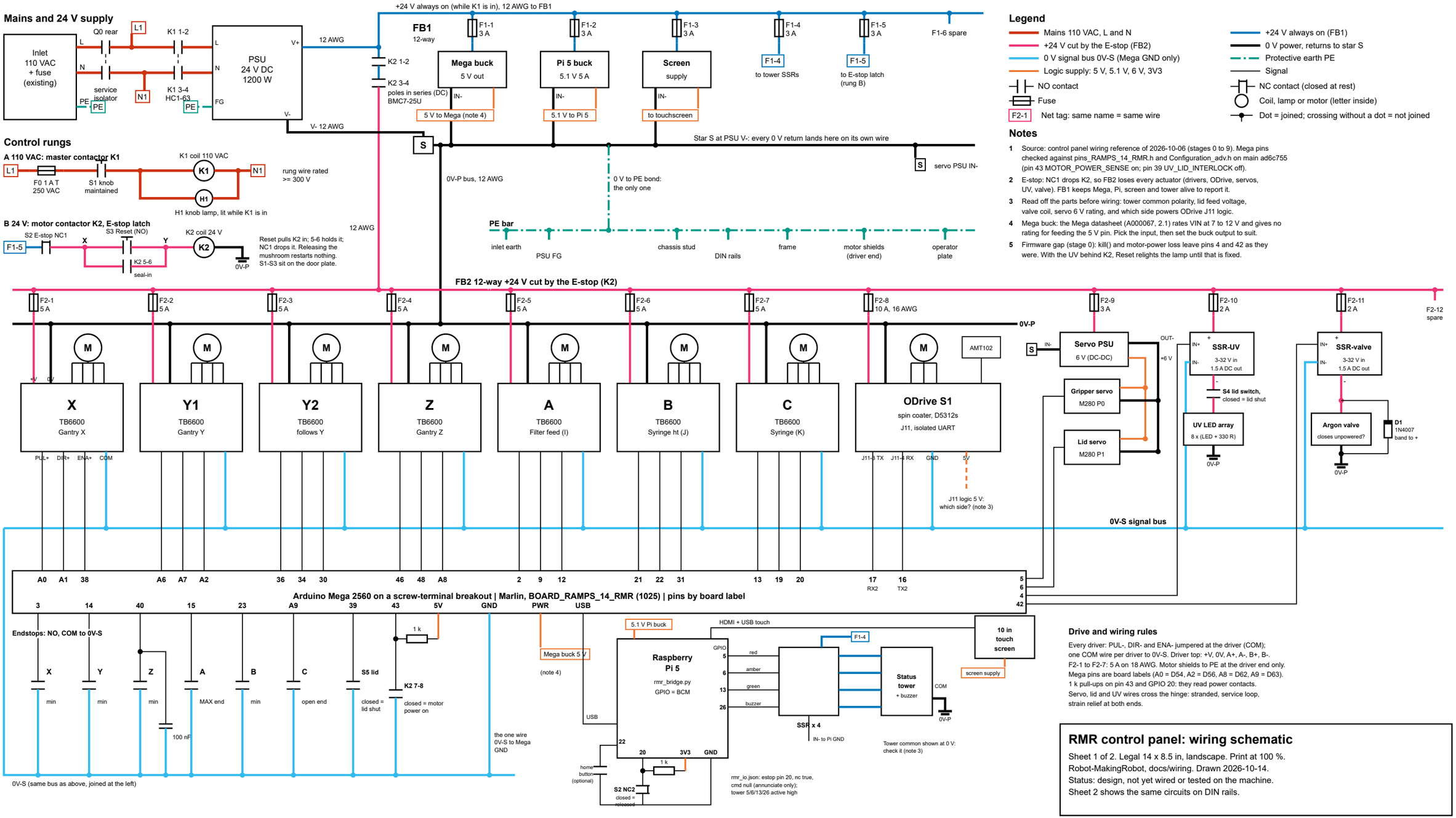
RMR control panel: wiring schematic

The layered E-stop panel as designed in the 2026-10-06 panel reference. Planned wiring, not the machine as it is wired today.



Legend

- Mains 110 VAC, L and N
- +24 V cut by the E-stop (FB2)
- 0 V signal bus 0V-S (Mega GND only)
- Logic supply: 5 V, 5.1 V, 6 V, 3V3
- NO contact
- Fuse
- F2-1 Net tag: same name = same wire
- +24 V always on (FB1)
- 0 V power, returns to star S
- Protective earth PE
- Signal
- NC contact (closed at rest)
- Coil, lamp or motor (letter inside)
- Dot = joined; crossing without a dot = not joined

Notes

1. Source: control panel wiring reference of 2026-10-06 (stages 0 to 9). Mega pins checked against pins_RAMPS_14_RMR.h and Configuration_adv.h on main ad6c755 (pin 43 MOTOR_POWER_SENSE on; pin 39 UV_LID_INTERLOCK off).
2. E-stop: NC1 drops K2, so FB2 loses every actuator (drivers, ODrive, servos, UV, valve). FB1 keeps Mega, Pi, screen and tower alive to report it.
3. Read off the parts before wiring: tower common polarity, lid feed voltage, valve coil, servo 6 V rating, and which side powers ODrive J11 logic.
4. Mega buck: the Mega datasheet (A000067, 2.1) rates VIN at 7 to 12 V and gives no rating for feeding the 5 V pin. Pick the input, then set the buck output to suit.
5. Firmware gap (stage 0): kill() and motor-power loss leave pins 4 and 42 as they were. With the UV behind K2, Reset relights the lamp until that is fixed.

Drive and wiring rules

Every driver: PUL-, DIR- and ENA- jumpered at the driver (COM); one COM wire per driver to 0V-S. Driver top: +V, 0V, A+, A-, B+, B-. F2-1 to F2-7: 5 A on 18 AWG. Motor shields to PE at the driver end only. Mega pins are board labels (A0 = D54, A2 = D56, A8 = D62, A9 = D63). 1 k pull-ups on pin 43 and GPIO 20: they read power contacts. Servo, lid and UV wires cross the hinge: stranded, service loop, strain relief at both ends.

RMR control panel: wiring schematic

Sheet 1 of 2. Legal 14 x 8.5 in, landscape. Print at 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-14.
Status: design, not yet wired or tested on the machine.
Sheet 2 shows the same circuits on DIN rails.